



A Systematic Review of Self-Regulated Learning through Integration of Multimodal Data and Artificial Intelligence

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Abstract

While behavioral, contextual, and physiological data streams have long been used to investigate self-regulated learning (SRL), a systematic understanding of the current state how different data streams and modalities contribute to measuring regulation processes across diverse learning contexts remains limited. This systematic literature review provides a foundational step toward this understanding by addressing two objectives: (1) identifying which data streams and modalities researchers have used to capture cognitive, affective, metacognitive, and motivational (CAMP) processes underlying SRL, and (2) examining how multimodal data analytics have been applied to capture the temporal and sequential characteristics of these processes across study contexts. The studies were mapped onto the Self-Regulated Learning Processes, Multimodal Data, and Analysis grid, a two-dimensional framework with CAMP processes on one axis and multimodal data streams on the other. By evaluating four analytic approaches—unimodal, horizontal, vertical, and integrated—we identify how different combinations of data streams and processes have been employed. Our findings reveal a shift from unimodal approaches (one data stream, one process), to more integrated approaches (combining multiple data streams and processes). Although multimodal data collection is increasingly common, gaps remain, especially in measuring motivation and affective states. Furthermore, analytic methods often do not reflect alignment between data streams, with standard statistics predominating even in integrated approaches, whereas AI-based analytics may be more suited. This review positions itself as a foundational step in advancing SRL measurement, offering insights into current practices and highlighting the need for more sophisticated methods to capture SRL across diverse learning contexts.

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Introduction

Given the rapid advancements in artificial intelligence (AI) and its significant impact on jobs (Eloundou et al., 2024), developing self-regulated learning (SRL) skills is essential for individuals to navigate these changes effectively and ensure lifelong learning. The field of SRL has a rich history of scholars across diverse disciplines investigating learners' engagement and regulation (Greene et al., 2024). Throughout this history, SRL has been characterized by its covert, cyclical, and recursive nature (e.g., Winne & Hadwin, 1998; Zimmerman, 1989). These characteristics have led researchers to increasingly operationalize SRL constructs with multimodal data and capture their recursive nature by applying advanced analytics (Järvelä & Bannert, 2021). Molenaar et al. (2023) introduced the so-called self-regulated learning processes, multimodal data, and analysis (SMA) grid to map development of SRL research visually. This grid mapped the authors' joint and individual research (63 papers) over the last five years along two axes: the CAMM processes (cognition, metacognition, affective states, and motivation) and common data streams (such as log files, eye tracking, and physiological data). Critically, four analytic approaches in these studies were defined to address how they used different process-oriented data stream(s) to measure CAMM processes.

Whereas Molenaar et al. (2023) focused on preselected authors and their work, this paper goes beyond and investigates all available research in a systematic review. The relevant SRL research was mapped according to their approach for investigating modalities through data streams in the SMA grid. Our main research question is: What multimodal data streams and analytical methods have been used in various learning contexts and environments to measure the cognitive, metacognitive, affective, and motivational processes and their interactions involved in self-regulated learning?

Advances in learning analytics (LA), cognitive sciences, and AI have provided the field with increasingly more sophisticated techniques and tools to better detect, diagnose, and act upon SRL (Azevedo, 2020; Bannert et al., 2017; Molenaar, 2022; Roll & Winne, 2015). Researchers collect various single and multimodal data streams to detect SRL strategies and actions, such as log data, physiological data, and eye tracking. They have attempted to translate these data into diagnosing CAMM processes using various analysis techniques, including process mining, discourse analysis and hidden Markov models. With these diagnoses, the aim is to make learning technologies more intelligent in terms of what, when and how to act upon the learners' struggles through personalised support such as student dashboards and scaffolding.

Although the SRL field continuously explores new ways to detect and diagnose SRL, there remains a need for a comprehensive and visually engaging synthesis of research conducted so far to discuss the interactions between the different components of SRL processes. First, we explain the two axes of the visual SMA grid:

CAMM processes and data streams. Second, the interactions between the two axes are discussed into the analytical approaches found previously in Molenaar et al.'s paper (2023).

Self-Regulated Learning Multimodal Data, Analysis Grid

The SMA grid consists of two axes that visualise how studies have detected and diagnosed SRL so far. SRL is a cyclical and recursive goal-oriented process in which students' cognition, affect, metacognition, and motivation are intertwined (Winne, 2015). These cognitive, affective, metacognitive, and motivational processes are commonly known as the CAMM processes (Azevedo, 2015). These form the x-axis of the SMA grid: (1) *Cognitive processes* (e.g. reading, elaboration, summarising) needed to study a topic; (2) *Metacognitive processes* (e.g., orientation, planning, goal setting, monitoring and evaluation), to actively monitor and control learning; (3) *Affective states* (e.g., boredom, confusion, frustration) to regulate emotions during learning; and (4) *Motivation* (e.g., learner's self-reported achievement goal orientation, interest, task value) to engage in an appropriate level of effort (Greene & Azevedo, 2010). External (e.g., task) and internal factors (e.g., knowledge and skills) play a role in how students enact SRL and these different CAMM processes (Efklides & Schwartz, 2024; Li et al., 2022; Winne & Hadwin, 1998). The dynamic interplay between the CAMM processes in conjunction with the internal and external factors is thought to be key for understanding how individuals become successful self-regulating learners.

SRL is about adjusting resources, goals, and strategies to optimize learning over time (Winne, 2017). Data collected during learning processes is needed to detect this inherent characteristic of SRL and ensure a clear theoretical alignment between data and SRL (Gašević et al., 2015; Wise & Shaffer, 2015). Thus, the y-axis of the SMA grid represents the different categories of online data streams. Azevedo (2015) made the distinction between process, product, self-report, and knowledge construction data when measuring different CAMM processes. He illustrated how the different data sources are ideal for capturing only some of the CAMM processes, highlighting the necessity for multimodal learning analytics, especially when focusing on multiple or even all the CAMM processes (see also Molenaar et al., 2023). The taxonomy proposed by Di Mitri et al. (2018) made the distinction between data about behaviour (including various verbal and non-verbal modalities, such as speech, gaze, and physiological measurements) and context (e.g., social and environmental factors). It was also highlighted how multimodal approaches allow modelling learning across physical and digital worlds, which is particularly relevant in current SRL research where AI and other digital tools play a pivotal role in practising, measuring, and supporting SRL. More recently, Sailer et al. (2024) categorized data sources based on their level of obtrusiveness, indicating that tests and self-reports are highly obtrusive whereas digital trace data are minimally obtrusive. Obtrusiveness refers to the extent to which data collection interferes with typical learning activities. Trace data, such as search behavior during an essay task, reflect natural student activity and require no additional tools or interruptions, unlike tests or self-reports, which

disrupt the learning process. While the taxonomies by Di Mitri et al. (2018) and Sailer et al. (2024) are valuable, they are insufficiently granular to account for the specific characteristics commonly theorised in SRL models.

In SRL research, three categories for the data streams have been distinguished (Azevedo, 2015). These categories are (1) *Behavioural data*—entailing data streams that detect aspects of students' behaviour during learning, reasoning, and problem-solving—which includes log files, keyboard and mouse data, eye tracking data, chat content, and screen recordings; (2) *physiological data*—indicating students' bodily responses during learning—in the form of electrodermal activity (EDA), heart rate variability (HRV), or electroencephalogram (EEG); and (3) *contextual data*—including data sources that enable to derive a meaningful inferences of students' behaviours or physiological responses (Schneider, 2021) when they interact with learning technology (e.g., intelligent tutoring system, serious games, immersive virtual environments) and other humans (e.g., parent, instructor, teachers, peer, artificial agent)—which includes self-reports, interviews, video data, voice data (think-aloud or dialogue). Although multimodal learning analytics can offer real-time insights into SRL using physiological or behavioural data, contextual data helps researchers to understand why specific CAMM processes occurred (or did not occur) in a given learning situation (Järvelä & Hadwin, 2024). When designing a study and selecting data streams, researchers' choices may also reflect considerations about the degree to which learners are aware of the data collection process and how this awareness might influence their SRL behaviours (McCardle & Hadwin, 2015). More obtrusive data streams—such as those associated with contextual data, including self-reports or think-aloud protocols—are likely to increase learners' metacognitive awareness of their actions. In addition, the inclusion of tools for monitoring or planning, which are captured in behavioural log data, may not only record but also shape learners' regulatory behaviours during the task.

Different analytic approaches were defined by Molenaar et al. (2023) to address the number of process-oriented data stream(s) that were used to diagnose one or multiple CAMM processes. The *unimodal approach* is a classical approach, focusing on a single CAMM process with one data stream. As Molenaar et al. (2023) described, four studies used log data exclusively to detect cognitive or metacognitive processes. For instance, Harley et al. (2018) investigated learner interactions with a pedagogical agent in MetaTutor, finding associations between goal-setting interactions and learning gains. The *horizontal approach* entails multiple CAMM processes measured with one data stream. Twenty-one studies explored (relationships between) multiple CAMM processes using a single data stream. One common horizontal data stream is voice data; for example, Molenaar and Chiu (2014) investigated students' discussions in triads to detect the sequential patterns between cognitive and metacognitive actions using statistical discourse analysis.

Additionally, the advancement of technology to collect multimodal data as well as the accessibility to learning technologies (e.g., intelligent tutor systems) has facilitated the integration of multimodal data streams in SRL research, and this is represented in the last two approaches. The *vertical approach* entails using multiple data streams to measure one CAMM process. Eleven studies focused on one SRL process measured by multiple data streams. For instance, Dindar et al. (2019) investigated

physiological synchrony and its association with metacognitive monitoring activities, revealing significant associations with learning outcomes. In a strict horizontal approach, researchers examine how different CAMM processes manifest in one data stream, while a strict vertical approach reveals how multiple data streams capture one CAMM process.

Lastly, we introduce the *integrated approach*, which entails multiple data streams aiming to capture multiple CAMM processes. This integrated approach is considered a wide category and can be a combination of the other approaches. For example, it could be a study combining a unimodal approach, capturing motivation through EDA, with a vertical approach for measuring cognition through eye tracking and log files to study how motivation and cognition intertwine. Another example is to align the data streams of eye tracking, think-aloud, log files and mouse and keyboard data to measure distinctive cognitive and metacognitive processes (Fan et al., 2023).

Analytical Methods

Azevedo and Gasevic (2019) have discussed various challenges when analysing such multimodal data, and resolving those requires expertise from educational psychology, data sciences, and LA fields (see also, Ochoa, 2022). An inherent challenge in this field is the variation in operational definitions of constructs when researchers aim to correlate latent constructs with observable, multimodal data (Gray & Bergner, 2022; Ochoa, 2022; Wiedbusch et al., 2023), making the synthesis of existing studies particularly difficult. Currently, there are two types of analytical methods that researchers have used when analysing multimodal data for studying CAMM processes. First, in the *sequential* method of analysing multimodal data, we mean that researchers analyse and model different data streams separately, one after the other, without temporal alignment of multimodal data and need to model multimodal data with different sampling frequencies. Therefore, this method enables contrasting the insights based on the different data streams even though the granularity and sampling frequency of the data streams differ (Azevedo & Gasevic, 2019). For example, Zhidkikh et al. (2023) analysed the Motivated Strategies for Learning Questionnaire (MSLQ) self-report survey and digital trace data separately and then cross-tabulated students' self-reported (MSLQ) and observed learning strategies from digital trace data.

Second, in the *parallel* method of analysing multimodal data, researchers model different data streams simultaneously. In this method, researchers may need to resolve challenges, including the temporal alignment of multimodal data with different sampling rates and handling missing or noisy data even from multiple data streams (Azevedo & Gasevic, 2019; Rovers et al., 2019). For example, Törmänen et al. (2023) identified different types of socio-emotional interactions in collaborative learning by conducting a mixture of hidden Markov modelling for video and electrodermal activity (EDA) data. This mixture simultaneously enabled capturing of regulation episodes and emotional valence based on video data and the emotional activation from the EDA data. When clustering by a mixture of hidden Markov

models, covariates (e.g., based on self-report data) can be used to explain cluster memberships facilitating the use of the parallel approach even in the presence of self-report data.

Even though researchers apply conventional, inferential statistics to get insights into the different CAMM processes (e.g., van Halem et al., 2020), current AI capabilities also enable reliance on human-AI synergies when pre-processing and analysing multimodal data (Järvelä et al., 2023). However, these increased capabilities also call for a critical evaluation of the amount of data and number of data streams needed to make inferences about CAMM processes of interest (Azevedo & Gasevic, 2019). Moreover, researchers need to be aware of how AI-enhanced data pre-processing and possible data transformations impact interpretations and inferences that can be made (Azevedo & Gasevic, 2019). Sailer et al. (2024) considered the ease of interpretation when categorizing different data processing and analysis techniques. For example, although the output of regression models and neural network models may have the same degree of complexity, explaining the results of regression models are easier, while neural networks, though more flexible, are more difficult to explain in a research paper. As an example, Suraworachet et al. (2024) applied and compared rule-based, machine learning, and generative AI approaches to identify cognitive, affective, motivational, or metacognitive challenges in face-to-face collaborative learning. They used AI-generated transcriptions of students' discourse as a basis for further analyses. To identify the different types of challenges, they extracted various features (e.g., dialogue act, sentiment, and emotions of different utterances) from the transcriptions using natural language processing. Their analyses relied on researchers' (humans') expertise on what features matter when detecting challenging moments from collaborative learning discourse. Relying on AI capabilities alongside humans' annotations sped up the analysis (over 10,000 utterances were eventually transcribed and analysed).

Integrating these data streams is complex and challenging and thus, may be time-consuming enough to cause issues when attempting to use actionable insights from the data to support learning in authentic settings (Azevedo & Gašević, 2019; Li et al., 2022; Ochoa, 2022). Despite these challenges, the field has matured enough so that many systematic reviews have preceded this paper and shed light on different aspects of SRL research.

Previous Systematic Reviews on SRL Research

The systematic review by Saint et al. (2022) included 53 empirical papers that focused on the temporality of the different SRL dimensions. By excluding the studies that only performed count-based statistical methods, they found that digital trace data and self-reports were the most common data sources, and more than half of the studies applied multiple data streams to measure the temporal aspects of SRL, with process mining, epistemic network analysis, and sequential analysis being the most typical analysis methods. Even though Saint et al. (2022) coded the theoretical models in the articles, they did not systematically differentiate which CAMM processes were the focus of the included studies.

Viberg et al. (2020) conducted a review of 54 empirical articles that applied LA to measure and support SRL in online learning environments. They found that most of the studies focus on measuring SRL in forethought and performance phases (Zimmerman, 2000), with a limited number of studies addressing the reflection phase. However, most studies were unclear about the SRL phases or CAMM processes they were measuring. Since they focused on SRL in online learning environments, the studies primarily relied on digital trace data, with most studies conducted in higher education. Generally, they found little evidence of the positive effects of LA on students' SRL, as most of the studies focused on measuring without supporting SRL.

Sharma et al. (2024) reviewed 59 empirical papers to study the measurement of self-regulated learning and socially shared regulation of learning in collaborative settings with adaptive learning technologies. They found that the different CAMM processes have received similar attention in the literature when learning collaboratively with adaptive learning technologies. However, they noted a predominance of survey data and a lack of utilizing multimodal data with more of a focus on the product (collaboration artefacts or learning outcomes) rather than process aspects.

These review articles share similar insights: most of the studies have been conducted in higher education contexts and have relied on digital trace and self-report data to measure (S)SRL (Noroozi et al., 2020; Saint et al., 2022; Sharma et al., 2024; Viberg et al., 2020). None of these review studies has explicitly mapped how different data streams and analytical methods can be used to measure the various CAMM processes of (S)SRL. Noroozi et al. (2020) did not limit their focus to (S)SRL; they reviewed 207 empirical studies and found that self-reports (surveys and interviews) mainly were used when applying multimodal data to study CAMM processes. In addition to digital trace data, audio and video data were the most used data streams. They also found that the studies most frequently focused on measuring (meta)cognitive processes with less attention given to affective and motivational processes. Even though the Noroozi et al. review did not elaborate on the approaches used to analyse the multimodal data streams, it showed integration of CAMM processes, as one-third of the included articles measured more than one CAMM process with multimodal data.

Despite recent systematic reviews on SRL research, there is no comprehensive synthesis that focuses on which combinations of data streams and analytical approaches have been used to measure different CAMM processes across learning contexts (e.g., individual and collaborative learning) and environments (e.g., online and in-classroom learning) from the SRL theory perspective.

Aims and Research Questions

The aim of the current study was to systematically investigate which different combinations of data streams and approaches (unimodal, horizontal, vertical, integrated) have been used in measuring learners' CAMM processes across various contexts and environments. Building on previous work by Molenaar et al. (2023), we aimed to provide a comprehensive and visual synthesis of the research conducted so far by addressing the following research questions (RQs):

RQ1: How have researchers employed various methodological approaches (unimodal, horizontal, vertical, and integrated) in detecting SRL CAMM processes, and what data streams have been utilized across these different detection frameworks?

RQ2: How do various data streams, both individually and in combination, align with the SMA grid framework to capture and represent CAMM processes?

The third research question pertained to the analytical methods (e.g., sequential or parallel methods, standard statistics) that accompany the unimodal and multimodal approaches. Given the advancements in machine learning and other types of sequential and parallel methods, we particularly anticipated using such methods in the vertical and integrated approaches. Moreover, we explored the specific analyses employed, such as clustering or process mining.

RQ3: How do the analytical methods differ across the four methodological approaches, and what distinctive characteristics emerge from their comparative analysis?

These first three questions map the typical combinations of data streams, analytical approaches, and CAMM processes. In addition to studying how researchers have evidenced the CAMM processes underlying SRL, we also aimed to explore contexts in which these studies have been conducted in terms of factors such as learning scenarios (individual or collaborative), study context (authentic or laboratory) and age of the students (Noroozi et al., 2020; Saint et al., 2022; Sharma et al., 2024; Viberg et al., 2020). This aim led to the following research question:

RQ4: How do the four methodological approaches differ in their application across various learning contexts, and what patterns emerge from this contextual analysis?

Our analyses delved into the challenges and opportunities of facilitating the valid and reliable measurement of CAMM processes using various data streams across diverse research designs and outcome measures. We identified areas of focus and underexplored research avenues for future investigation. To delve deeper into the variation in operational definitions of CAMM processes when applying multimodal data and analytics (Gray & Bergner, 2022; Ochoa, 2022), our study also provides an open-source database (<https://zenodo.org/records/15481052>). This database can be utilized to gain more detailed information about the SRL constructs within each CAMM process and the related indicators derived from multimodal data and analytics.

Methods

We followed the preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P; Shamseer et al., 2015) when searching for and selecting the included studies. The search and selection process was composed of four main steps: (1) identification, (2) screening, (3) eligibility, and (4) inclusion. In the following section, we elaborate on these steps individually.

We used three databases (ERIC, Scopus, Web of Science) to identify peer-reviewed empirical journal articles and proceedings published in English. All searches were conducted between November 2022 and January 2023. For the identification, we used the search terms referring to (a) self-regulated learning, (b) multimodal data, and (c) analytics (see Table 1). We used the AND and OR operators to identify the articles. Depending on the options provided by the database, the search terms also had to appear somewhere other than in the main text (i.e., in the title, abstract, or keywords). We identified 9416 articles and exported them to Rayyan (Ouzzani et al., 2016) for the initial screening of abstracts. Rayyan was also used to remove the 461 duplicates automatically.

We screened the titles and abstracts of these 9416 articles without accessing their full text. We decided to include an article, if all the following criteria were fulfilled:

1. It is an empirical journal or proceedings article.
2. It is published in English.
3. It focuses on studying self-regulated learning (SRL) or socially shared regulation of learning (SSRL) and related cognitive, affective, motivational, and metacognitive (CMM) processes in individual and collaborative learning settings.

Table 1 The search terms used to identify the articles relating to the (a) self-regulated learning, (b) multimodal data, and (c) analytics

Main construct	Search terms
Self-regulated learning	TITLE-ABS-KEY (srl OR "*"regulat*") OR TITLE- ABS-KEY ((learn* OR teach* OR train* OR school OR educat* OR student) AND (cognit* OR metacognit* OR motivation* OR affect* OR emotion*)) AND
Multimodal data	TITLE-ABS-KEY (multimodal* OR multichannel OR triangulat* OR"eye*tracking"OR gaze OR dialogue OR"think*aloud"OR interview OR"screen*capture"OR"screen*recording"video OR facial OR voice OR audio OR log OR trace OR mouse OR keyboard OR physiologic* OR"heart*rate"OR"electrodermal*activity"OR eda OR"skin*conductance"OR"galvanic skin response"OR"skin temperature"OR"electromyography"OR emg OR electrocardiography OR ecg OR electroencephalography OR eeg) AND
Analytics	TITLE-ABS-KEY (process OR sequence* OR sequential OR temporal* OR analytic* OR mining OR "network analysis" OR "signal process*" OR "artificial intelligence" OR "AI" OR "natural language process*" OR "NLP" OR "machine learning" OR "deep learning" OR "deep network*" OR "neural network*")

Note. We used the AND operator to identify the articles that include all constructs

4. It includes process data from single or multiple data streams. We define process data as a data source capturing the sequence and timing of events, actions, states or behaviours during a learning task.
5. It applies process-oriented analysis or analytical approaches.
6. The reported study is conducted in formal education (including laboratory studies).

An article was excluded if at least one of these criteria was not met. For this initial screening, among a team of eight coders, the 9467 articles were divided, and the titles and abstracts were screened for inclusion or exclusion. Two coders then recoded a sample of 25 articles from each coder, resulting in a total of 200 double-coded screenings. Cohen's kappa indicated that there was an 89% agreement between the screeners ($\kappa = 0.66$). Disagreements were resolved afterwards in the joint meetings among the authors. In this phase, we excluded 9209 articles.

The same team of eight coders assessed the eligibility of 258 articles, basing decisions on the full texts. Here, we excluded 116 articles for one or more of the following reasons: (a) No process-oriented analysis or analytical approaches applied ($N = 65$); (b) No process data collected ($N = 44$); (c) Focus is not on (S) SRL and related CAMM processes ($N = 32$); (d) Not an empirical journal or proceedings article ($N = 12$); (e) No full text is available ($N = 4$); (f) Context is not in formal education ($N = 4$). Thus, we included 142 articles for further analysis. Cohen's kappa indicated that there was an 88% agreement between the screeners ($\kappa = 0.66$) in this phase. Disagreements were resolved through negotiations in joint meetings among the coders.

To address RQ1 on the prevalence of the different approaches, we coded which CAMM processes (i.e., cognition, metacognition, affective state, motivation) were diagnosed as well as the common data streams divided into the categories of behavioural, contextual, and physiological data (see Appendix Table 1 for all codes). The approaches (unimodal, horizontal, vertical, or integrated) were coded for each study by calculating the number of CAMM processes measured as well as the number of data streams used. For example, the CAMM processes (1) metacognition and (2) cognition measured by one data stream, namely log files, would be coded as a horizontal approach. Studies using one data stream to measure one CAMM process would be defined as using an unimodal approach; studies using multiple data streams for one CAMM process are coded as a vertical approach. Finally, studies analysing multiple CAMM processes with multiple data streams are categorized as integrated approaches intended for complex groupings. The frequency of diagnosed CAMM processes in the studies, as well as the percentage prevalence of these approaches across the publication year, is reported.

To address RQ2, which combinations of data streams were used to capture one or more relationships between CAMM processes, we mapped the studies onto the SMA grid. The frequency of studies investigating the same combination of modalities and data streams is visualised in the grid to show typical combinations. Furthermore, to answer RQ3, the analytical methods were coded and categorised into qualitative analysis, clustering, Markov modelling, process mining, standard statistics, (lag) sequential analysis (lag seq. analysis), network analysis, decision trees,

Support Vector Machine (SVM), Natural Language Processing (NLP), Recurrence Quantification Analysis (RQA), and discourse analysis (Appendix Table 1).

Lastly, to answer RQ4, we coded the research design and learning context details, including (1) sample size (i.e. $N = 1\text{--}50$ participants; $N = 51\text{--}100$; $N = 101$ and more; these three intervals were chosen so that those would differentiate between various analytical approaches, the number of intervals would not be too high to ensure the ENA remained interpretable, and each sample size range would contain a sufficient number of studies), (2) whether conducted in authentic or laboratory learning environments, (3) data collected in one single session or across multiple sessions, (4) educational level of participants (i.e., primary, secondary, or higher education), and (5) individual or collaborative learning settings (Appendix, Table 1). To analyse the similarities and differences among various analytical approaches in terms of study characteristics in RQ4, we employed epistemic network analysis (Shaffer et al., 2016) to our data using the ENA Web Tool (version 1.7.0) (Marquart et al., 2018). The ENAs helped to visualize the relationships between the target CAMM processes, collected data streams, applied analysis methods, and research design details for each analytical approach. In the epistemic networks, the previously mentioned coded items served as nodes, and the width of the edges between the nodes was proportional to the frequency of articles in which both respective coded items were present (the scale for the edge width was the same (weight = 2) for all the plots except for the difference networks where it was doubled to highlight the differences between the ENA plots). To facilitate the interpretation of the networks, we positioned the nodes into the unit circle, spacing them equally. This was done instead of optimising the node positions that minimise the difference between the plotted points and their corresponding network centroids, which would have made the visual interpretation of the networks more challenging. To make the comparison of different analytical approaches with varying numbers of articles meaningful, we weighted group networks by their mean, not their sum. This decision to use the mean also enabled us to plot and interpret the differences between the epistemic networks for the different analytical approaches.

Results

Prevalence of Unimodal, Horizontal, Vertical, Integrated Approach Over Time

Starting from the year 2000, SRL research has primarily focused on capturing evolving cognitive (36.3%) and metacognitive processes (44.5%), as shown in the distribution height of these processes on the left side of Fig. 1. Since 2007, there has also been a smaller but increasing interest in diagnosing motivation (10.2%) and affective states (9.0%). The data streams reported across publication years are predominantly in the behavioural and contextual data categories. In recent years, physiological data have also been used in SRL research, but they still represent only a small portion of the field.

When ranking the most common online data streams used across all years, we found log data as the most frequently used source ($N = 79$, 39.7%), followed by

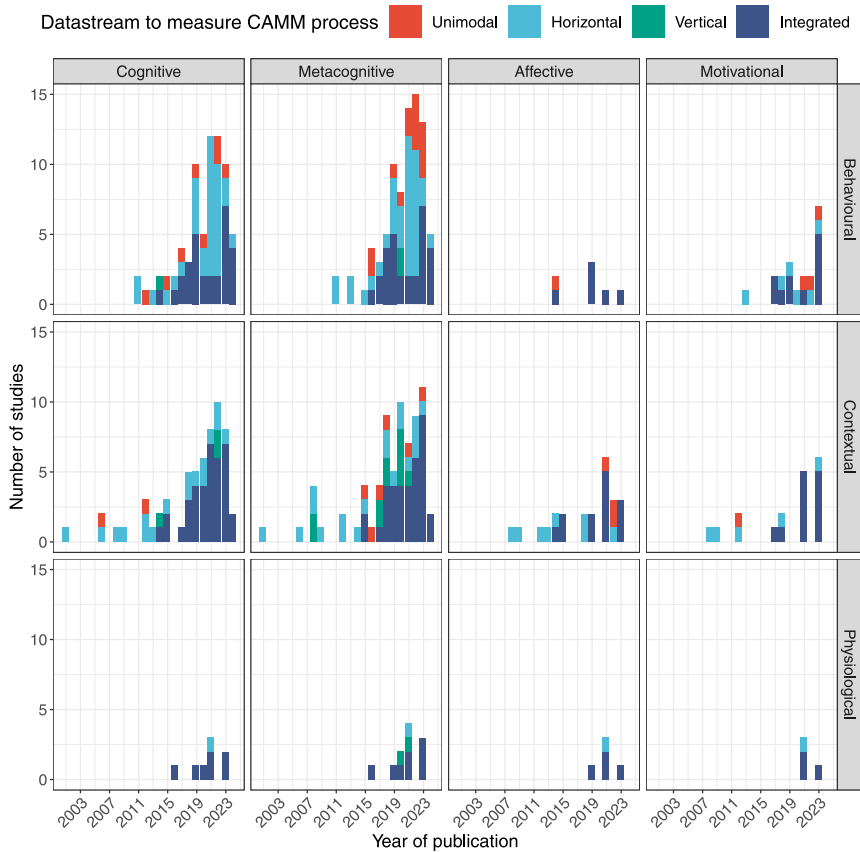


Fig. 1 Historical development of CAMM process measurement (cognitive, metacognitive, affective and motivational) across data categories (i.e., behavioural, contextual and physiological data)

think-aloud ($N = 24$, 12.1%), video data ($N = 23$, 11.6%), audio dialogue ($N = 20$, 10.1%), self-reports ($N = 16$, 8.0%), EDA ($N = 10$, 5.0%), eye tracking ($N = 8$, 4.0%), chat content ($N = 7$, 3.5%), interviews ($N = 4$, 2.0%), mouse and keyboard data ($N = 4$, 2.0%), HR(V) ($N = 2$, 1.0%), pen strokes ($N = 1$, 0.5%), EEG ($N = 1$, 0.5%). For the historical development of individual data streams, please see Appendix Fig. 1.

There is a distinct historical development in the approaches to the number of CAMM processes used with the number of data streams collected. When aggregating publication years 2010 to 2016, we found that 11 studies used an unimodal approach (37.9% of the total across these years), focusing on one CAMM process detected by one data stream. In comparison, for publication years 2017–2023, there was a noticeable decline in the percentage of unimodal approaches ($N = 11$, 25.4%). Most studies continued to use the horizontal approach (i.e., two or more CAMM processes detected by one data stream), with a stable percentage rate going from 12 studies (41.4%) in earlier years to 41 studies (40.2%) more recently. Fewer studies used a vertical approach (i.e., one CAMM process detected by two or more

data streams), with only 2 studies (6.9%) in earlier years and 6 studies (5.9%) more recently.

Figure 1 shows in more detail that the horizontal approach was frequently used to diagnose cognitive and metacognitive processes using behavioural data. More recently, there was also an increase in the horizontal approach to motivational research. Studies with a vertical approach tended to focus especially on metacognitive processes through contextual data, and in the last two years, there were no studies with a vertical approach. A significant development across the years is the increased use of an integrated approach in SRL research, growing from 4 studies (13.8%) in earlier years to 29 studies (28.4%) more recently. As shown in Fig. 1, this integrated approach increased for all CAMM processes and data categories.

Combination of Data Streams to Diagnose CAMM Processes

Mapping the studies onto the SMA grid reveals various combinations of data streams used to diagnose CAMM processes and their relationships (Fig. 2). The most prevalent combination is the use of log data to investigate cognitive and metacognitive processes through a horizontal approach ($N = 26$ studies, 18.8%; e.g., Taub et al., 2022). Unimodal approaches using log data to examine cognition ($N = 5$ studies, 3.6%; e.g., Li et al., 2021) and metacognition ($N = 13$ studies, 9.4%; e.g., Kim et al., 2018) are also common. Additionally, log data are employed to study metacognitive and motivational processes in a horizontal approach ($N = 4$ studies, 2.9%). For instance, Chung and Hsiao (2019) examined how study persistence and an entropy-based value of study regularity influenced course performance using clustering methods, although they did not consider the relationship between cognitive and metacognitive processes. In the vertical approach, studies utilized a variety of data streams, primarily focusing on metacognitive processes. Malmberg et al. (2019), for example, measured joint monitoring processes during a collaborative exam. They first used video and think-aloud data to qualitatively code critical moments of monitoring activity. Subsequently, they measured the frequency of peak synchrony in physiological arousal using EDA at these moments. However, no clear alignment was found between monitoring events and physiological synchrony.

The integrated approach is visualised in various combinations of data streams and modalities. There were three studies that considered all CAMM processes using log data and self-reports in a sequential manner. Khat and Vogel (2022), for instance, studied students' self-reports of their self-regulated learning behaviours with trace data from a self-regulated learning management system (LMS). They used standard statistics to investigate whether perceptions from self-reports were associated with log actions in the LMS. Another sequential integrated approach is demonstrated by Zheng et al. (2019), who combined chat content and log data from high school and higher education students engaged in a STEM task. The authors found that cognitive elaboration and executive SRL actions could be derived from log data, while planning and task analysis could only be inferred from students' chat content. This approach allowed them to investigate both cognitive and metacognitive processes and their sequential patterns through sequential mining of both data streams. An

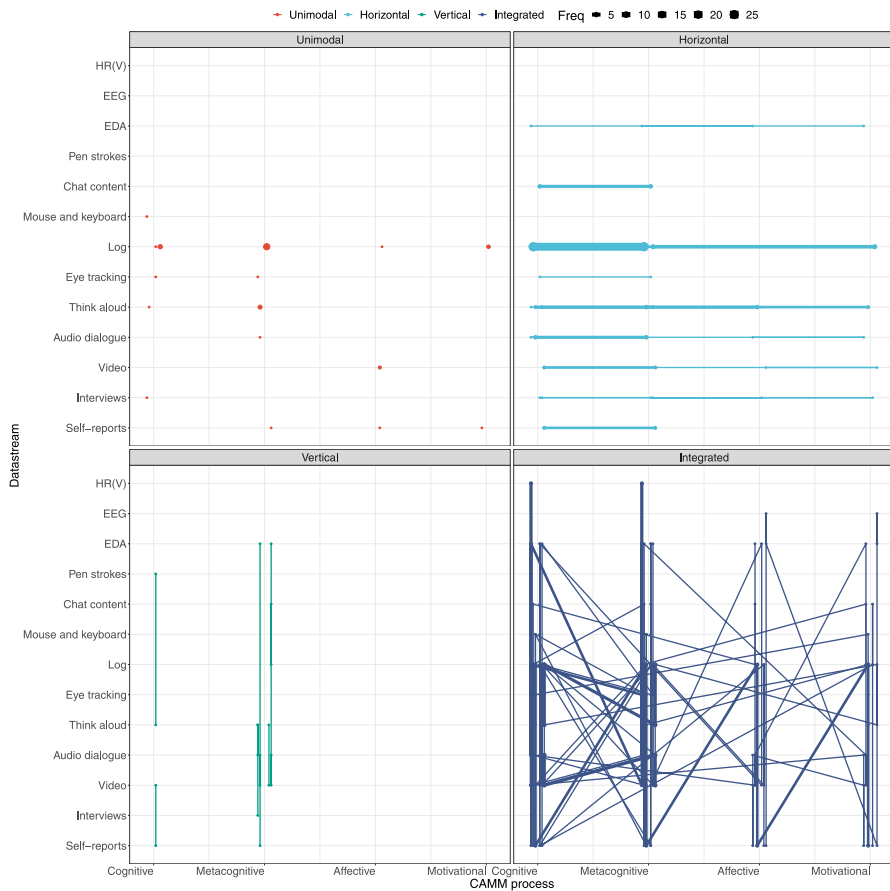


Fig. 2 SMA (Self-regulated learning, Multimodal, Analysis) grid mapping of 141 studies. The x-axis represents CAMM processes, and the y-axis the data collected during learning. Lines and dots visualize combinations of CAMM processes and data streams, with thickness indicating frequency. Colours denote approach types: red (unimodal), light blue (horizontal), green (vertical), and dark blue (integrated)

example of an integrated approach where data streams were analysed with a parallel method to validate trace data is the work of Fan et al. (2023). During an essay writing task, they collected trace and think-aloud data from higher education students in the lab. They aligned the two data streams on a timeline and put the matched and unmatched occurrences of the same SRL processes detected in the data streams into pMineR, a process mining analytical technique.

Differences in Analytical Methods in the Four Approaches

We assessed the analytical methods used in the research by applying the different approaches. Only analytical methods mentioned in three or more studies were included in our analysis. This criterion excluded studies using decision trees ($N=$

1,0.7%), RQA ($N = 1,0.7\%$), SVM ($N = 1,0.7\%$), and NLP ($N = 2,1.4\%$), Fig. 3 illustrates the most common analytical methods, with descriptive statistics being the dominant category ($N = 74$, 53.6%) across years and approaches. Notably, this prevalence of standard statistics is also observed in the integrated approach ($N = 19$ studies, 13.8%), where studies often examine different processes and data modalities sequentially, rather than synchronizing data streams (e.g., examining process A with modality X, then process B with modality Y), as exemplified by Khiat and Vogel (2022). In the vertical approach, descriptive statistics was used in earlier years, but data streams are now integrated through either clustering, network analysis, process mining or Markov modelling in recent years.

Process mining is the second most employed method ($N = 19$), particularly in integrated and horizontal approaches in recent years. This method is beneficial for analysing learner interactions with learning platforms, specifically sequences of learner actions. It is primarily applied to log data, as demonstrated in studies by

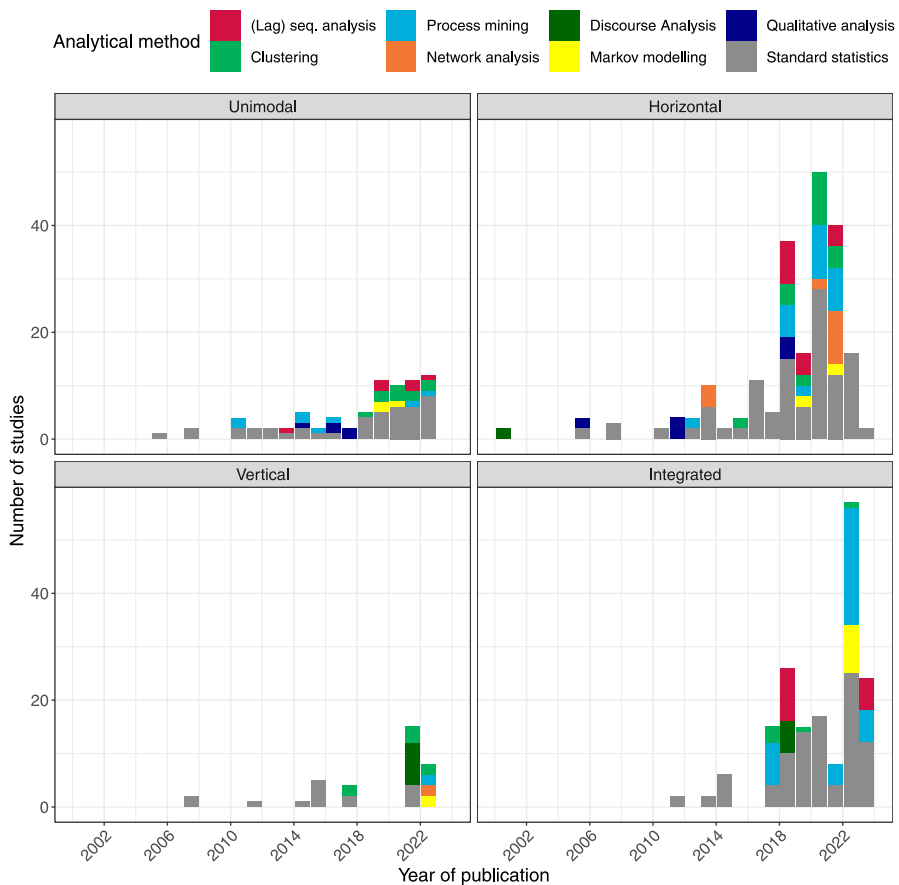


Fig. 3 Categories of analytical methods used in the unimodal, horizontal, vertical and integrated approach across the publication years

Zheng et al. (2019), Maldonado-Mahauad et al. (2018), and Wong et al. (2021). Similarly, lag sequential analysis in the integrated approach, network analysis in the horizontal approach, and discourse analysis in the vertical and integrated approach are commonly used methods for investigating connections between learning processes. Figure 3 shows that although inferential statistics are still widely used, other methods that account for SRL characteristics (e.g., process mining, sequential analysis and Markov modelling) have become more common in recent years especially in the horizontal and integrated approaches.

Researchers often combine multiple analytical methods. For instance, Matcha et al. (2020) exemplify a common combination: first applying Markov modelling to extract distinctive patterns in sequences of learning actions, followed by clustering techniques to categorize learners into strategy groups. In their study, this approach yielded four clusters related to motivation that were associated with personality traits (as measured with Big Five).

Epistemic Networks of Factors Represented in the Four Approaches

In the following section, we present summaries of each analytical approach (unimodal, horizontal, vertical, and integrated) based on the epistemic networks of the included articles. In the epistemic networks, nodes represent the codes from the included articles. The size of the node represents the frequency of that code in the included studies. The stronger the edges between two nodes, the more frequently these two codes co-occurred in the included articles within a particular analytical approach. First, Fig. 4a illustrates how log or think-aloud data, in conjunction with standard statistical analysis, have been utilized to measure mostly metacognitive processes when researchers have adopted an unimodal approach. These studies typically feature relatively large sample sizes (with many studies involving more than 100 participants) and originate from authentic higher education learning environments where participants have conducted individual learning tasks over multiple sessions.

Second, Fig. 4b demonstrates how log data has been analysed by applying standard statistical methods, clustering and process mining to measure cognitive and metacognitive processes in the horizontal approach. These studies share similar characteristics with those in the unimodal approach, particularly concerning the focus on individual learning tasks over multiple sessions in authentic higher education learning environments. Moreover, a significant number of these studies have been conducted with more than 100 participants.

Third, Fig. 4c highlights the triangulation of data streams, such as think-aloud, audio, or EDA, and log data, when researchers aim to capture primarily metacognitive processes with multiple data streams in the vertical approach. This diversity of data streams is also evident in the methods applied, as these studies have utilized RQA, Markov modelling, other sequential analysis approaches, NLP techniques, and inferential statistical analysis. However, as the vertical approach was relatively rare among the included studies, the application of these different methods is present only in a few papers. When applying the vertical approach, the focus was more

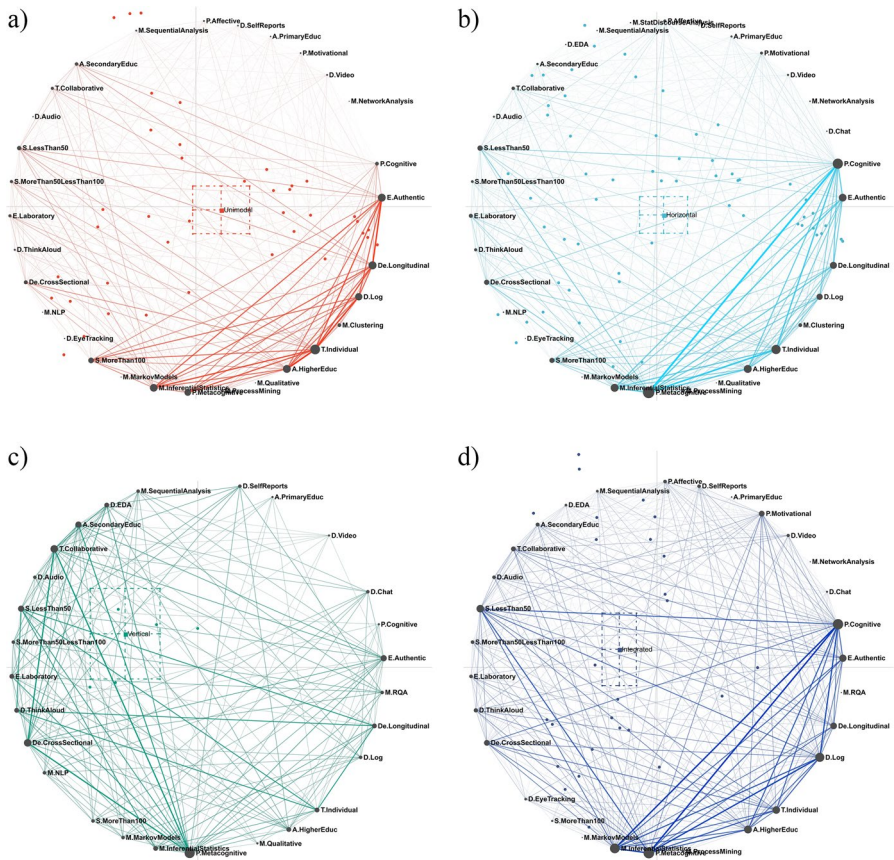


Fig. 4 Epistemic Networks visualized according to different analytical approaches: **a)** unimodal, **b)** horizontal, **c)** vertical, and **d)** integrated. Each network depicts the following data streams: Data Nodes (D), CAMM processes (P), analytical methods (M), research design (De), sample size (S), learning environment (E), age group (A), and task type (T). The dots within the networks represent individual articles. The means (represented by squares) and confidence intervals (represented by dashed lines) of the networks are also depicted

frequently on collaborative learning tasks in secondary education than on individual learning tasks in higher education.

Fourth, Fig. 4d shows how log data, complemented with other data streams, form the basis for measuring primarily cognitive and metacognitive processes in the integrated approach, similar to the horizontal approach. In the integrated approach, the data sources were predominantly analyzed by applying inferential statistical analysis to measure these processes. Even though most of the studies have been conducted in authentic learning environments with relatively small sample sizes (less than 50 participants), they have involved both individual and collaborative learning tasks over single or multiple sessions in secondary and higher education.

To summarize the comparison of these approaches, the unimodal and horizontal approaches predominantly rely on large sample sizes and focus on individual learning in higher education, whereas the vertical approach often uses smaller samples and diverse data streams for collaborative learning tasks in secondary education. The integrated approach merges elements from both horizontal and vertical approaches, complementing log data with other data sources in both individual and collaborative learning tasks, yet typically involves fewer participants. In all these approaches, studies were more frequently conducted in authentic learning environments than in laboratories.

Discussion

The overall aim of the systematic review was to identify challenges and opportunities in facilitating valid and reliable measurement of CAMM processes using various data streams across diverse research designs and outcome measures while highlighting areas for future investigation in the complex field of SRL.

High Prevalence of Horizontal and Emerging Integrated Approach

Regarding the prevalence of different approaches and data streams (RQ1), we found that horizontal approaches are the most dominant through the years. At the same time, there is also a clear developmental shift from studies using unimodal approaches towards more integrated, multimodal approaches (for examples of the integrated multimodal approaches, see also, Järvelä & Bannert, 2021). Unlike Noroozi et al. (2020), who identified survey data as predominant, this review found log data and think-aloud protocols to be the most common data streams. This discrepancy likely stems from our focus on process-oriented analyses during learning, whereas surveys are typically administered before or after learning.

Mapping of CAMM Processes in the Grid, Underexploration of Motivation

As was also found by Molenaar et al. (2023), our analysis revealed that metacognition and cognition were the most studied CAMM processes, while especially motivation received less attention, especially when using multimodal data streams. Motivation as a construct is such that it is difficult to measure with multimodal measures. It has its origin in learners' belief systems and, thus, has a multifaceted nature that interacts with various psychological, social, and environmental factors to drive behaviour (Efklides & Schwartz, 2024; McInerney & Etten, 2001). Instead, connecting traditional self-report measures to multimodal methods may be a way to measure motivational and emotional conditions, their situational and contextual variations and dynamic regulation to SRL (Järvenoja et al., 2023).

Patterns in Approaches and Dominance of Standard Statistics Use

To answer RQ2, we visualised key patterns in the SMA grid. First, log data were predominantly used to investigate cognitive and metacognitive processes through horizontal approaches. Unimodal approaches using log data to study either cognition or metacognition independently were also common. While less frequent, vertical approaches employed diverse data streams, particularly for studying metacognitive processes. In the integrated approach, though rare, there were attempts to capture all CAMM processes using multiple data streams. However, challenges for both approaches in modelling data streams simultaneously were evident. This challenge also arose when examining analytical methods (RQ3), where our review revealed that standard statistics remained the dominant analytical method across all approaches, including in the integrated approach. Related to the review of Saint et al. (2022), process mining emerged as the second most common method, especially in integrated and horizontal approaches in recent years, when analysing multiple data streams. In learning analytics studies involving data from MOOCs in general (Zhu et al., 2022), we also see this trend where descriptive statistics is most used, followed by machine learning. Significant variations in study contexts across the four approaches (RQ4) underscore the importance of considering contextual factors when interpreting SRL research findings. Unimodal and horizontal approaches typically involved large samples (over 100 participants) in authentic higher education settings, focusing on individual tasks over multiple sessions. The vertical approach showed more diversity, including secondary education settings, smaller samples (less than 50 participants), and a mix of individual and collaborative tasks. Integrated approaches, while mostly conducted in authentic settings with smaller samples, demonstrated the most diversity in terms of education level, task type, and session duration. The prevalence of particular approaches and data streams, mapping these to specific CAMM processes, the analytical methods employed, and the influence of study contexts provide valuable insights for researchers seeking to advance SRL measurement and understanding.

Alignment in Multimodal Approach

Our systematic review reveals a complex landscape of analytical methods in SRL research, reflecting both challenges and opportunities in analysing multimodal data to study CAMM processes, as Azevedo and Gasevic (2019) already pointed out. Even though more studies are conducting multimodal integrated approaches in data collection, the analysis used to handle these data does not reflect the complexity when trying to validate data streams or results. Studies focus more on individual processes and employ a sequential method of analysis instead of trying to align the temporality of data streams well to capture the dynamic relationship between data streams and modalities, in line with the parallel method. Aligning data streams is much more challenging to do and was rarely found in our review. Knowledge of advanced machine learning techniques that align and model CAMM modalities across different granularities is needed. In a systematic review that was done on

multimodal data in learning analytics (Mu et al., 2020), the authors distinguished the relationship of “many-to-one” (multimodal data vs one learning indicator). They explained that the advantage of such an approach lies in improving measurement accuracy and information richness and that the efficiency of the algorithm model needs to be determined by comparing different machine learning algorithms. Similarly, in the vertical approach, focusing on one CAMM process with multiple data streams, we can calculate this accuracy and information richness for specific modalities. A good methodological example that future research in SRL can take is the heterogeneous tripartite network approach, such as the non-related SRL work of Feng et al. (2024), where the interconnections between spatial movement, video and audio data were analysed to find individual and collaborative engagement patterns. Conversely, the horizontal approach allows for investigating relationships between CAMM processes using a validated data stream, focusing on the dynamic interplay between modalities. An interesting move forward in this approach is examples of human-AI synergies, such as the work of Suraworachet et al. (2024), and Borchers et al. (2024), who both transcribed think-aloud data through whisper.ai,¹ an open-source automatic speech recognition (ASR) system, and then human-coded it into several SRL processes.

Ideally, vertical and horizontal approaches can be combined into an integrated approach that captures all the complexity of SRL processes. Integrating data enhances information richness, aligning with the ‘many-to-many’ principle described by Mu et al. (2020), which also connects multimodal data with multiple learning indicators. An interesting angle to this is the work of Mejia-Domenzain et al. (2024) with their ‘multimodal approach’. They first applied a horizontal analytical approach using trace data and self-reports separately to cluster students according to their metacognitive and cognitive processes. Second, they applied a vertical approach by calculating the alignment rate between the clusters from the trace data and the self-reports with separate modalities, showing close alignment for metacognition, time management, and elaboration, with a slight agreement for effort. Thirdly, they used an integrated approach to investigate the interrelations between the data streams of both modalities using a Canonical Correlation Analysis. This novel multimodal approach provides a more nuanced understanding of how data streams contribute to investigating SRL modalities.

Bridging Theory, Context, and Methodological Complexity

Our review revealed significant differences in analytical approaches across educational levels, individual versus collaborative settings, and laboratory versus classroom contexts. Some approaches and data streams appear more appropriate for individual lab settings with higher education students than for other contexts. Consistent with Viberg et al. (2020), we found that most studies were conducted with higher education students using digital trace data. Transferring results from one context to

¹ <https://openai.com/research/whisper>

another remains a challenge, but work like that of Lämsä et al. (2025) demonstrates promising approaches. By building on laboratory studies from Fan et al. (2022, 2023) using an integrated approach to align think-aloud and trace data with higher education students to diagnose metacognitive and cognitive processes, Lämsä et al. (2025) successfully applied a trace-based engine to reveal cognitive and metacognitive processes in secondary education classroom settings.

A key issue raised using multimodal data in learning research is how such data can be meaningfully interpreted through existing theoretical frameworks. For example, Baker and Reiman (2024) point out that many theories in educational psychology—commonly used to understand collaborative learning—were not developed with high-temporal-granularity data in mind. These theories are typically directed at explaining at a higher granularity level and do not explain or predict fine-grained behavioural data, such as event sequences, but instead focus on changes in aggregated variables. This limits their capacity to account for the situated nature of learning processes and social interaction, favouring abstractions that detach behaviour from its context. Moreover, they are designed to be tested using statistical models based on linear relationships, which are ill-suited for capturing the feedback loops and complexities inherent in dynamic systems. Järvelä and Hadwin (2024) have pointed out a risk that epistemological chasms may arise if researchers operationalise theoretical constructs inductively from multimodal data without thoroughly understanding self-regulated learning as a phenomenon needed to guide data analytic decisions deductively. In response to that, they introduce a theory-driven trigger regulation framework for advancing multimodal analytical approaches to research self-regulated learning.

Limitations and Future Directions

Several limitations of our study should be noted. We could not include SRL models in our analysis due to the prevalence of multiple models reported in 23% of the studies, which prevented their use in the Epistemic Network Analysis. We did include the coded SRL models in the open-source database. We noticed a lack of clarity on the theoretical lens that was used in the papers to conceptualise and measure SRL processes, a finding that the review of Sharma et al. (2024) on collaborative learning also pointed out. There is a need to analyse the different operationalization of CAMM processes as an addition to the grid to fully reveal how comparable different approaches in fact are at a micro-analysis level. Such a detailed analysis could help the field increase conceptual clarity and may drive more advanced theoretical models. Additionally, the large number of coders involved in this extensive systematic review may have contributed to a relatively low inter-rater agreement ($\kappa=0.66$). Future studies might benefit from limiting the size of coder teams to enhance consistency.

Moving forward, the field of SRL research needs to address several key challenges. First, future studies should aim to bridge the gap between theoretical models of SRL and empirical measurements, ensuring that data collection and analysis methods align with our understanding of SRL that CAMM processes

influence each other over time, which can be done using a horizontal or integrated approach (Greene et al., 2024). In addition, an evaluation of which methods or approach were more or less suited to answer the intended research questions in these studies could potentially be addressed with the dataset made available. In the future, there is a need to delve deeper into the variation in operational definitions of CAMM processes when applying multimodal data and multimodal learning analytics. Second, there is a need for more advanced analytical techniques, particularly those leveraging the power of AI, to exploit the potential of multimodal data fully. Critically, researchers should explore more parallel analytical methods to capture the temporal nature of SRL processes across different granularities using a vertical and integrated approach. Third, an area for future research is extending the SMA grid to incorporate the "act" part of the detect-diagnose-act framework (Molenaar, 2022), extending the grid with an axis. This would involve a systematic review of exploring how the insights from (multimodal) data streams to diagnose one or multiple CAMM processes are used to support SRL processes effectively. In addition, given that support interventions can be given before, during, and after the learning process, it would be vital to distinguish these three phases to detect new effective avenues for SRL support. Even though we did not explicitly focus on SRL support in our review, the reliable and valid measurement of SRL that the studies in this systematic review show is an essential basis for providing meaningful support.

In conclusion, our review reveals an interesting shift towards collecting more diverse data streams to capture multiple modalities of SRL by placing the studies into the SMA grid. This trend reflects the growing recognition of SRL's complex, multifaceted nature. However, our findings also highlight a gap between data collection and analysis methods. Many studies still rely on traditional descriptive statistics for analysing multimodal data, which is time-consuming to collect and, in most cases, inappropriately used due to violations of statistical assumptions (e.g., normal distribution). Moreover, not all studies use multimodal data for triangulation or validation measures and instead analyse the data streams separately. For example, Taub and Azevedo (2019) studied the effects of prior knowledge on cognitive and metacognitive SRL processes by first focusing on the eye tracking data and then on the log data. We recommend that researchers utilize AI capabilities to perform analyses that contribute to the validation of the measurement of CAMM processes with multimodal data streams. Additionally, automating the processing and analysis of multimodal data with AI will help scale up studies in this field (Di Mitri et al., 2018; Järvelä et al., 2023). By mapping current research approaches and identifying areas for improvement, this systematic review aims to guide future SRL research towards more comprehensive and nuanced methodologies.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10648-025-10028-0>.

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Data Availability The dataset supporting the findings of this study is publicly available and includes the coding data of the academic papers analyzed in this research. It can be accessed at: <https://zenodo.org/records/15481052>. All relevant codes, descriptions, and metadata are included to enable replication and further analysis.

Declaration

Conflict of interest The authors declare that they have no conflict of interest.

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